5 Fig. 5.1 shows two containers, each filled with hot water.

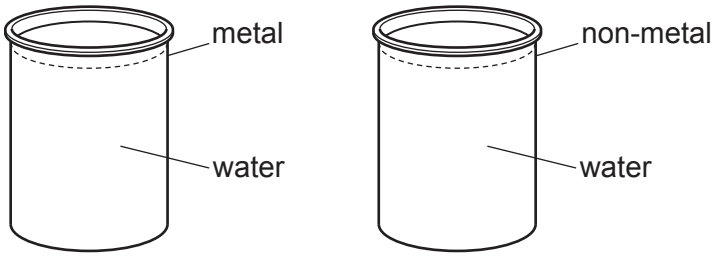


Fig. 5.1

The outer surface of the metal container is hot.

(a) Explain how electrons transfer thermal energy through the metal of the container.

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..... [3]

(b) The outer surface of the non-metal container is much cooler than the outer surface of the metal container.

Explain why a non-metal conducts thermal energy less well than a metal.

.....

..... [1]

(c) Explain, in terms of particles, why gases are poor thermal conductors compared to non-metal solids.

.....

.....

..... [2]

[Total: 6]

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